

6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotary encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is an ESP32-C6 module driving a full-color LCD screen. The ESP32 features wifi and bluetooth allowing for automatic time setting and smart home integration. Communication with the STM32 is via UART.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relays on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interface with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

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Sheet: /

File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

Size: A4

Date: 2025-12-01

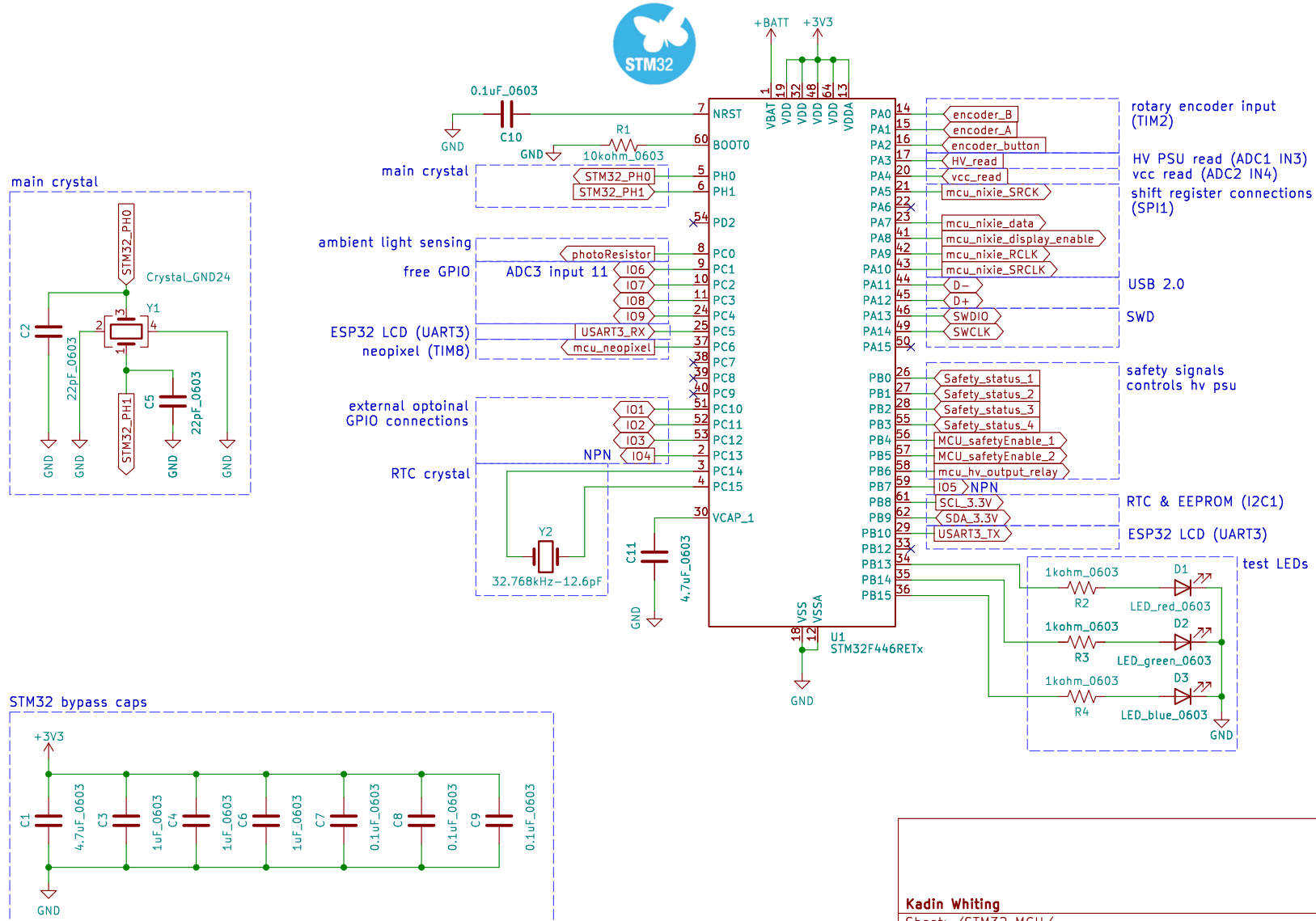
Rev: 1

KiCad E.D.A. 9.0.4

Id: 1/9

STM32F446 Processor ARM M4
168MHz

Level Shifting



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Sheet: /STM32 MCU/

File: STM32 MCU.kicad_sch

Title: Microcontroller

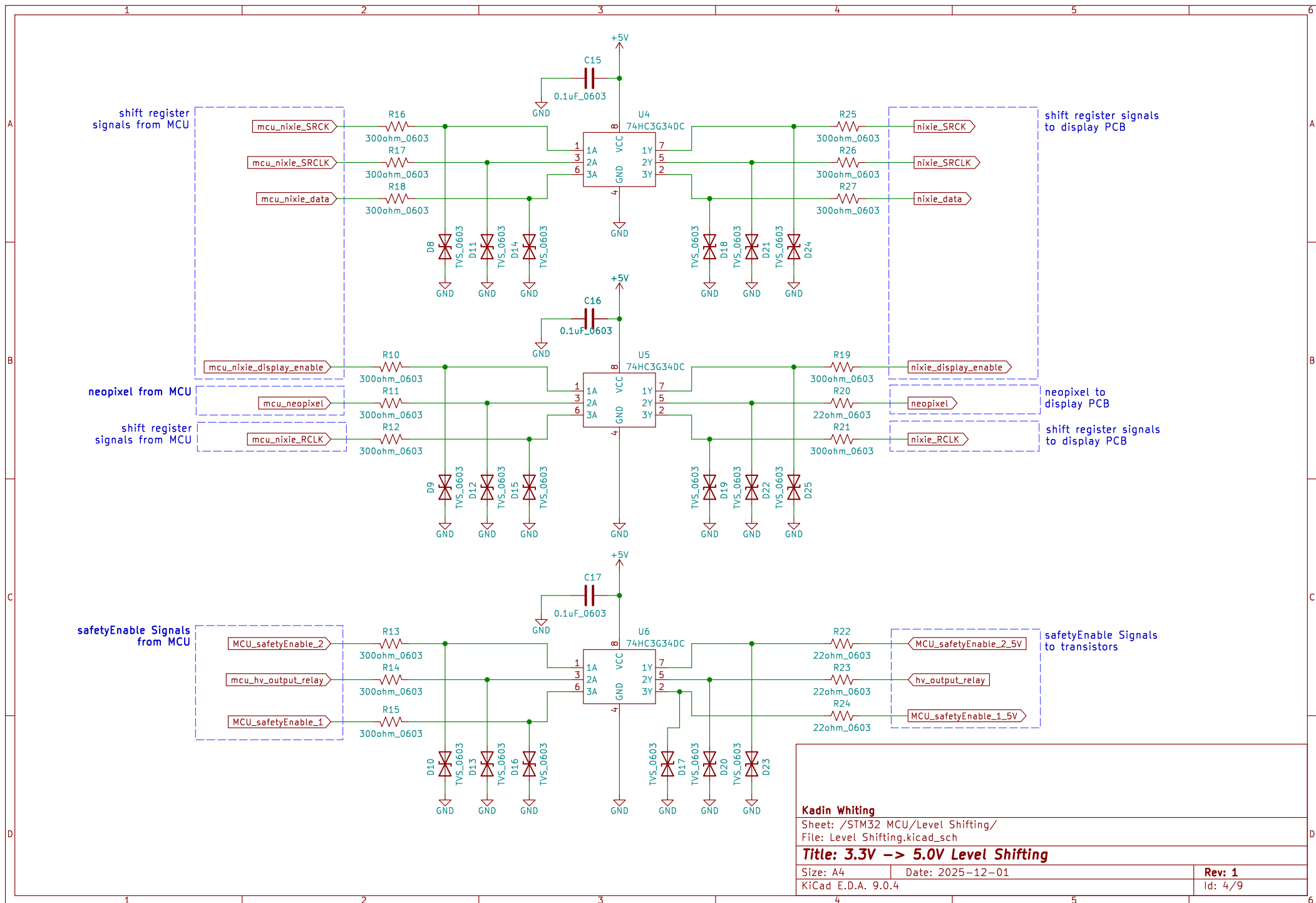
Size: A4

Date: 2025-12-01

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Id: 2/9



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Sheet: /STM32 MCU/Level Shifting/
File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

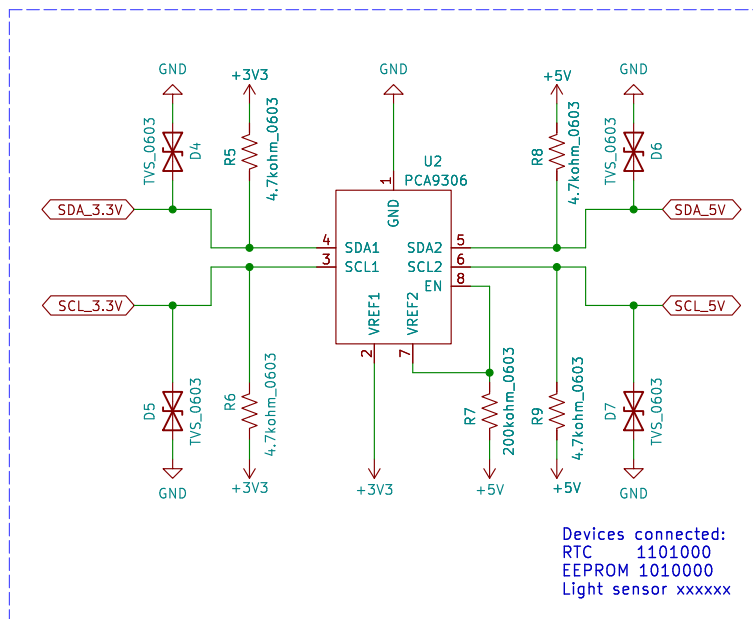
Size: A4 Date: 2025-12-01

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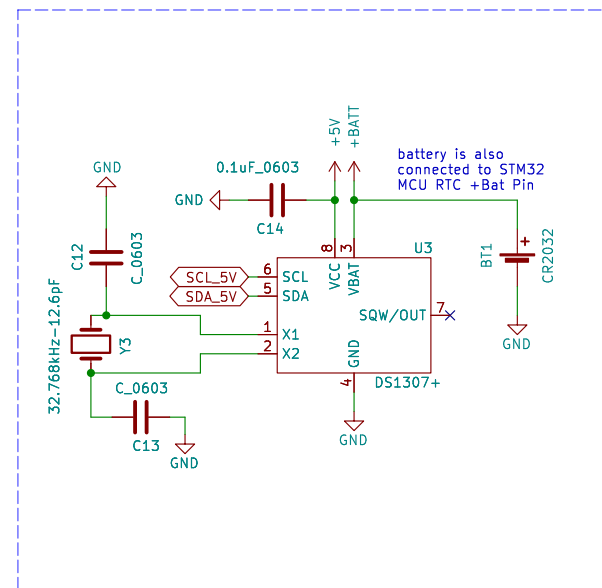
Rev: 1

Id: 4/9

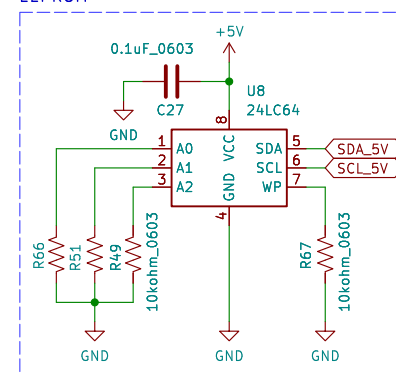
3.3V to 5V I2C level shifter



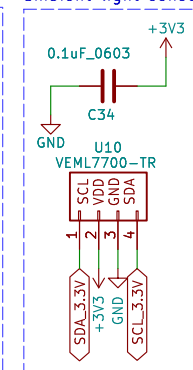
Real Time Clock



EEPROM



ambient light sensor



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Sheet: /I2C/

File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

Size: A4

Date: 2025-12-01

Rev: 1

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Id: 3/9

right-angle encoder 24 pulses/rotation
integrated momentary button
low pass filters included

The diagram shows a circuit for the photoresistor. A 10kohm_0603 resistor (R81) is connected between GND and a node. This node is also connected to a photoresistor (R80, R_Photo) and a +3V3 supply.

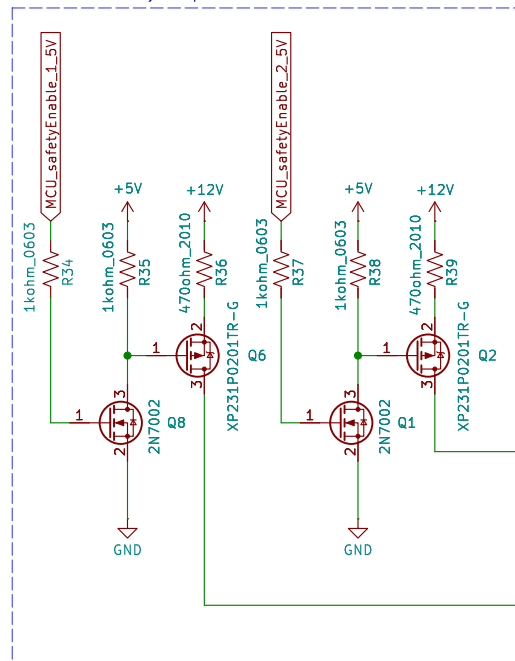
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File: Interface.kicad_sch

Size: A4	Date: 2025-12-01
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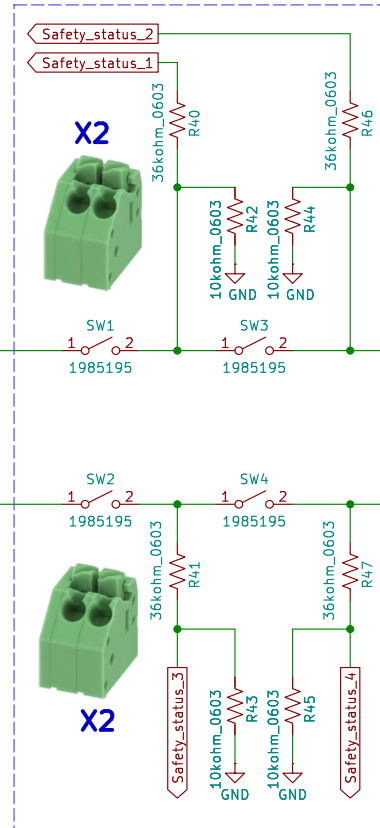
Rev: 1

Id: 5/9

two outputs from MCU
to enable safety loop



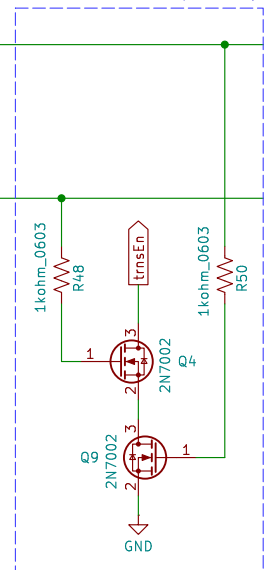
safety contacts for sensing if the lid is installed
feedback to MCU for each contact



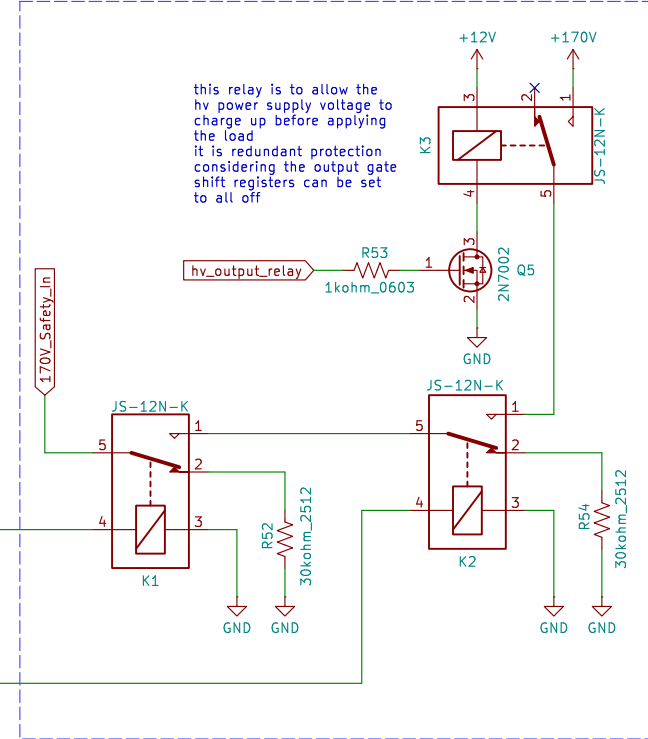
loop through
display PCB



enables HV PSU input relay



controls output of HV power supply



this relay is to allow the
hv power supply voltage to
charge up before applying
the load
it is redundant protection
considering the output gate
shift registers can be set
to all off

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Sheet: /Safety Circuit/

File: Safety Circuit.kicad_sch

Title: Safety Circuit – High Voltage PSU Control

Size: A4

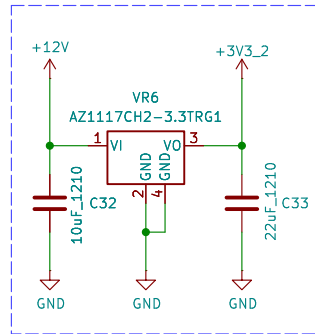
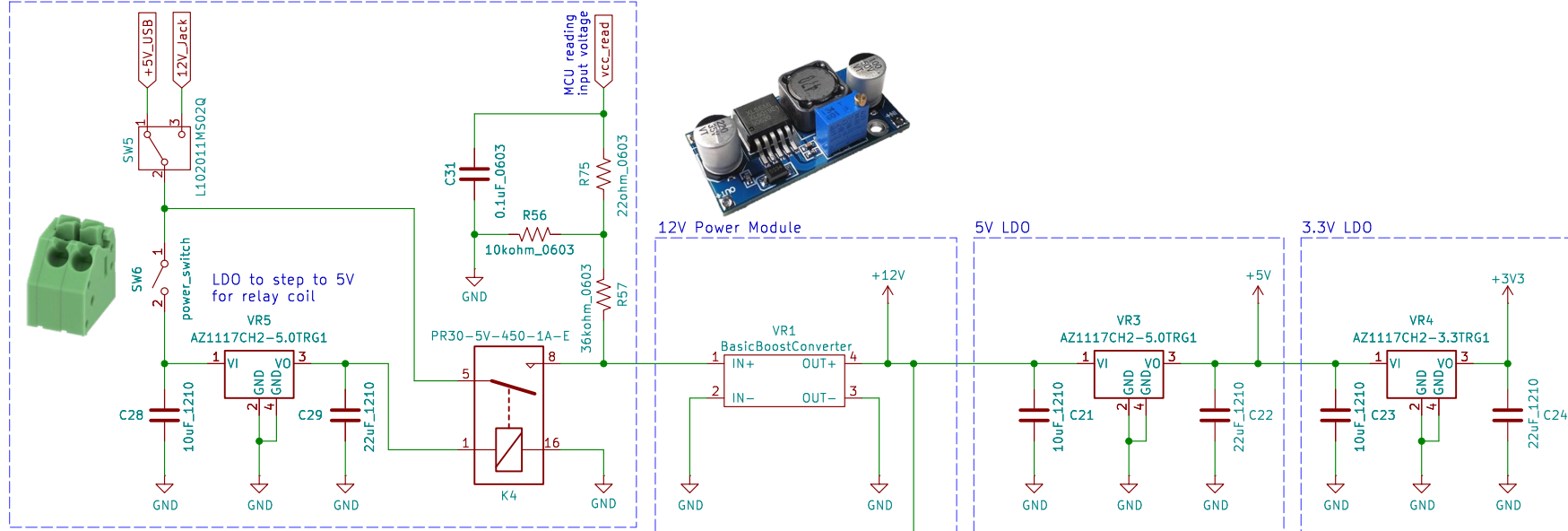
Date: 2025-12-01

Rev: 1

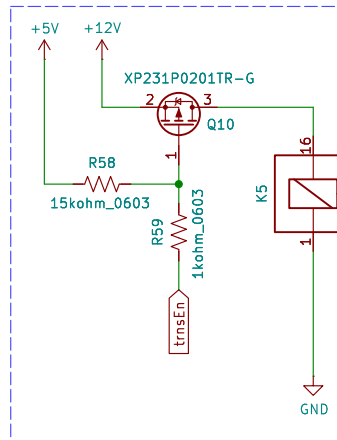
KiCad E.D.A. 9.0.4

Id: 6/9

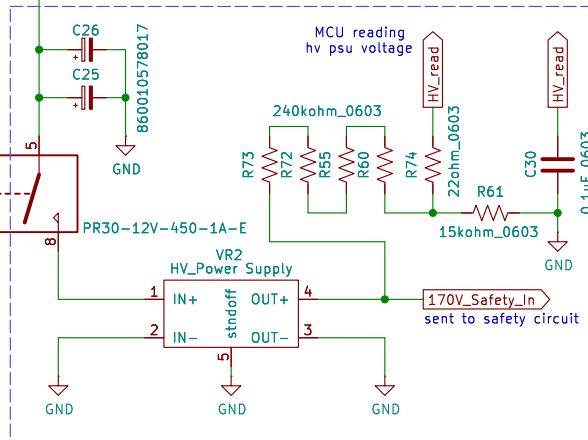
power selection and power switch



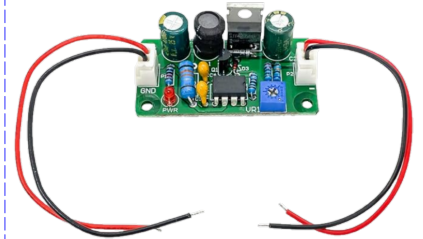
3.3V LDO - external devices



Controlled by safety circuit



high voltage power supply



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Sheet: /Power/
File: Power.kicad_sch

Title: Power

Size: A4 Date: 2025-12-01
KiCad E.D.A. 9.0.4

Rev: 1
Id: 7/9

The schematic diagram illustrates the SWDIO and SWCLK signal lines. The SWDIO line is connected to pin 1 of connector P1 (Conn_01x02) and pin 2 of connector J4 (M20-7920242R). The SWCLK line is connected to pin 3 of connector J4. Both lines are protected by TVS diodes (TVS_0603) and signal diodes (D30, D32) to ground. A USB dongle (Q7: XP231P0201TR-G) is shown connected to the SWCLK line. A +3V3 supply is also indicated.

Diagram illustrating the connection of a 12V Jack to the breadboard. The 12V Jack is connected to the positive rail (1) and the ground rail (2) of the breadboard. The ground rail is connected to the GND pin of the breadboard.

A schematic diagram of a 5-pin header. It shows a vertical bus of five green circular nodes. The bottom node is connected to a ground symbol labeled 'GND'. Each of the five nodes is also connected to a red circular pad labeled 'H1' through 'H5' respectively, with the text 'MountingHole_Pad' below each pad. The pads are arranged in a vertical column to the right of the bus nodes.

Diagram illustrating the wiring for a 12V Jack (J7) connected to a PJ-102A module.

Polarity: Inner: Positive +, Outer: Negative -

Wiring Schematic: The 12V Jack is connected to the J7 connector. Pin 1 (Green) connects to Pin 1 of the J7. Pin 2 (Blue) connects to GND. Pin 3 (Red) connects to Pin 3 of the J7, which then connects to GND.

Physical View: A black 12V Jack with a silver center pin and a black outer shell.

SCHEMATIC	
Model	PJ-102A
Center Pin	Ø2.0 mm

The image displays three wiring diagrams for 10-pin headers, labeled J1, J2, and J3, and a 3x10-pin header component.

J1 Wiring: A 10-pin header (61301021821) is connected to a 4-wire cable. The connections are: Pin 1 to nixie_data, Pin 3 to nixie_SRCLK, Pin 5 to nixie_SRCLK, Pin 7 to nixie_RCLK, Pin 9 to nixie_display_enable, Pin 2 to SafetyLoop_display1_out, Pin 4 to GND, Pin 6 to GND, Pin 8 to GND, and Pin 10 to GND.

J2 Wiring: A 10-pin header (61301021821) is connected to a 4-wire cable. The connections are: Pin 1 to +170V, Pin 3 to SafetyLoop_display1_in, Pin 5 to GND, Pin 7 to SafetyLoop_display2_in, Pin 9 to GND, Pin 2 to GND, Pin 4 to GND, Pin 6 to GND, Pin 8 to GND, and Pin 10 to GND.

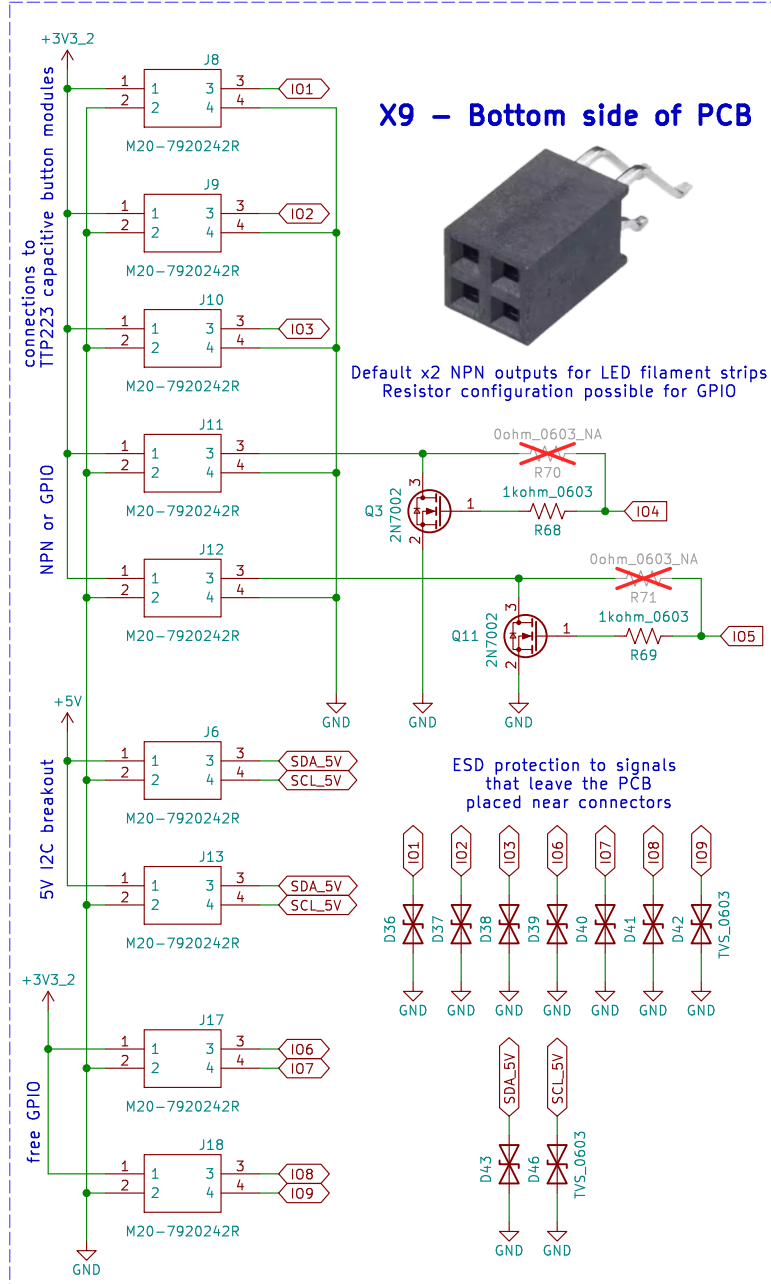
J3 Wiring: A 10-pin header (61301021821) is connected to a 4-wire cable. The connections are: Pin 1 to +5V, Pin 3 to neopixel, Pin 5 to GND, Pin 7 to GND, Pin 9 to GND, Pin 2 to SafetyLoop_display2_out, Pin 4 to GND, Pin 6 to GND, Pin 8 to GND, and Pin 10 to GND.

Header Component: A 3x10-pin header component is shown, labeled X3.

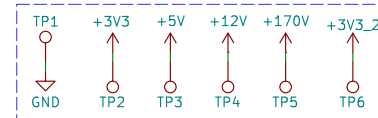
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Rev: 1
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external GPIO optional connections



power test points



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Sheet: /Connectors 2/

File: Connectors 2.kicad_sch

Title: Connectors

Size: A4

Date: 2025-12-01

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Rev: 1

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